

Multiple Choice (10 marks)

1. If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all $a, b \in G$, then G is a/an
 2. commutative semi group
 3. abelian group
 4. non-abelian group
 5. None of these

Find the order of the factor group $\mathbb{Z}_6 / \langle 3 \rangle$.

- (a) 2
- (b) 3
- (c) 6
- (d) 12

If $A = \{1, 2, 3\}$ then relation $S = \{(1, 1), (2, 2)\}$ is

- (a) symmetric only
- (b) anti-symmetric only
- (c) both symmetric and anti-symmetric
- (d) an equivalence relation

Let A be the set of all ordered pairs of integers (m, n) such that $7m + 12n = 22$. What is the greatest negative number in the set $B = \{m + n : (m, n) \in A\}$?

- (a) -5
- (b) -4
- (c) -3
- (d) -2

In the xy -plane, the curve with parametric equations $x = \cos t$ and $y = \sin t$, $0 \leq t \leq \pi$, has length

- (a) 3
- (b) π
- (c) 3π
- (d) $3/2$

True / False (10 marks)

Answer True or False

6. True or False: The product of 12 and 16 in the ring $\mathbb{Z}_{24}$ is congruent to 4.
7. True or False: An ideal N in a ring R with unity is all of R if and only if N contains some element other than 1.
8. True or False: The order of the factor group aH is not necessarily a divisor of $|a|$ in group theory.
9. True or False: The degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$ is 6.
10. True or False: The map $x \rightarrow axa^2$ of a group G into itself is a homomorphism if and only if G is abelian.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. There is an unlimited supply of congruent equilateral triangles made of colored paper. Each triangle is a solid color with the same color on both sides of the paper. A large equilateral triangle is constructed from four of these paper triangles as shown. Two large triangles are considered distinguishable if it is not possible to place one on the other, using translations, rotations, and/or reflections, so that their corresponding small triangles are of the same color. Given that there are six different colors of triangles from which to choose, how many distinguishable large equilateral triangles can be constructed? `[asy] draw((0,0)--(10,0)--(5,8.7)--cycle); draw((5,0)--(2.5,4.3)--(7.5,4.3)--cycle); [/asy]`
12. Find all values of x such that

$$\begin{vmatrix} 1 & 1 & 1 \\ x & 7 & -2 \\ x^3 & 343 & -8 \end{vmatrix} = 0.$$

Enter your values separated by commas. For example, if you think the possible values are 4, 5, and 6, then enter your answer as "4,5,6".

End of Paper